HERIOT-WATT UNIVERSITY

SCHOOL OF MATHEMATICAL AND COMPUTER SCIENCES

DEPARTMENT OF ACTUARIAL MATHEMATICS AND STATISTICS

F72ZA1
FINANCIAL MATHEMATICS 1

10 December 2004

Time allowed 2 hours

Attempt all questions

Total Marks 60

Only approved calculators may be used.

NUMERICAL SOLUTIONS

1. Fill in the missing values in the following table:

i	v	d	$i^{(3)}$	$d^{(2)}$
0.052632	0.95	0.05	0.051734	0.050641

2. (a) £90.21
   (b) £107.75

3. (a) 44.501
   (b) 16.385
   (c) 3.506

4. (a) Bookwork
   (b) (i) £53,649.15
   (ii) £53,673.37

5. (a) (i) £332.14
   (ii) £7,055.78
   (b) (i) £237.99
   (ii) £187.43

6. £64.98

7. (a) Verify that 0.05 is a solution of the equation of value.
   (b) The yield in 2003–4 is $i_1 = 0.06$. The yield in 2005 is $i_2 = 0.019715$. Hence LIRR is given as the solution of:

$$(1 + \text{LIRR})^3 = (1 + i_1)^2(1 + i_2)^1 \Rightarrow \text{LIRR} = 0.0464.$$

   (c) TWRR is given by:

$$\left(\frac{12,000}{11,000}.\frac{13,420}{13,000}.\frac{8,586}{8,420}\right)^{1/3} - 1 = 0.0472$$

8. (a) Yield is 10.0 % p.a.
   (b) Yield is approximately 11.3 % p.a.
   (c) Yield is 12.0 % p.a.

   Hence you would prefer C.